

BPC Training

Module 3 – Multivariate Analysis & MSPM System

9th – 11th Jan 2023
CC-SISPI FTKKP UMP

Topics

1. Multivariate Analysis

- Multivariate techniques & matrix operation
- Eigen-structure
- Variance-covariance transformation

2. MSPM System

- General framework of MSPM
- MSPM development – phase I
- MSPM development – phase II

Multivariate Analysis

Multivariate Techniques & Matrix Operation

Types of MVA Techniques:

1. Single criterion, multiple predictor association.

$$y = \alpha x_1 + \beta x_2 + \gamma x_3$$

LR

2. Multiple criterion, multiple predictor association.

$$\begin{aligned} y_a &= \alpha_a x_1 + \beta_a x_2 + \gamma_a x_3 \\ y_b &= \alpha_b x_1 + \beta_b x_2 + \gamma_b x_3 \end{aligned}$$

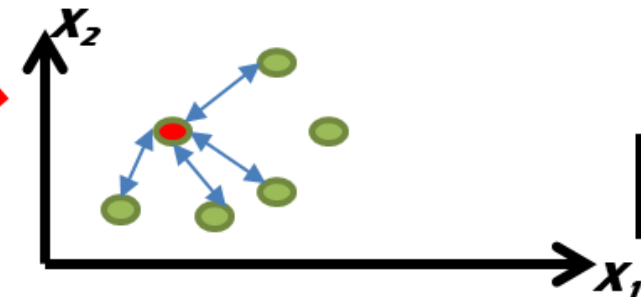
MLR

3. Analysis of variable interdependence.

$$PC_1 = \alpha_1 x_1 + \beta_2 x_2 + \gamma_3 x_3 + \delta_4 y_a + \varepsilon_5 y_b$$

PCA

4. Analysis of inter-object similarity.



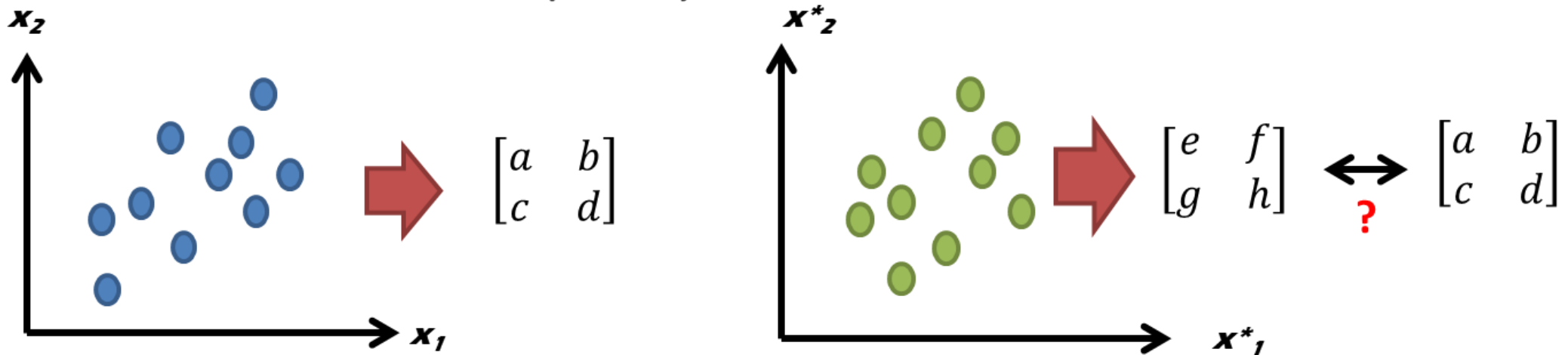
MDS

Multivariate Analysis

Multivariate Techniques & Matrix Operation

- **Objectives:** MVA is generally concerned with **association among multiple variables**.

Objective 1: Discover regularities on the behavior of 2 or more variables => **exploratory**



- **Benefits:** The 'variation' of a given population/system can be explained statistically.

Multivariate Analysis

Multivariate Techniques & Matrix Operation

Illustrative Data Matrix

$$\begin{array}{c}
 \text{Variables – by column} \\
 V_1 \quad V_2 \quad \dots \quad V_m \\
 \mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,m} \\ x_{2,1} & x_{2,2} & \dots & x_{2,m} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & x_{n,2} & \dots & x_{n,m} \end{bmatrix} \begin{array}{l} s_1 \\ s_2 \\ \cdot \\ \cdot \\ s_n \end{array}
 \end{array}
 \begin{array}{l}
 \text{Objects/samples} \\
 \text{– by row}
 \end{array}$$

Multivariate Analysis

Multivariate Techniques & Matrix Operation

Basic Matrix Operation

1. Minor product moment:

➤ $B_n = A'A$

Suppose,

A=

1	3
2	5
6	4

So, $B_n = A' * A$

Bn=

1	2	6
3	5	4

X

1	3
2	5
6	4

Bn=

41	37
37	50

2. Major product moment:

➤ $B_j = AA'$

Suppose,

A=

1	3
2	5
6	4

So, $B_j = A * A'$

$B_j =$

1	3
2	5
6	4

X

1	2	6
3	5	4

$B_j =$

10	17	18
17	29	32
18	32	52

Multivariate Analysis

Multivariate Techniques & Matrix Operation

Basic Matrix Operation

3. The mean corrected sums of squares and cross products (SSCP) matrix:

➤ $S = A_d' A_d$

Where A_d = mean corrected matrix

Suppose,
A =

1	3
2	5
6	4

Average
3 4

Then, $A_d = A - \text{Ave Matrix}$

$A_d =$

1	3
2	5
6	4

–

3	4
3	4
3	4

$A_d =$

-2	-1
-1	1
3	0

So, $S = A_d' A_d$

$S =$

-2	-1	3
-1	1	0

×

-2	-1
-1	1
3	0

$S =$

14	1
1	2

4. The covariance matrix:

➤ $C = [1/(n-1)](A_d' A_d)$

$A_d =$

-2	-1
-1	1
3	0

So, $C = [1/(n-1)]S$

$C =$

1	14	1
2	1	2

×

14	1
1	2

where, $n = \text{samples} = 3$

$C =$

7	0.5
0.5	1

Multivariate Analysis

Multivariate Techniques & Matrix Operation

Basic Matrix Operation

5. The correlation matrix:

$$\mathbf{K} = [1/(n-1)](\mathbf{A}_s' \mathbf{A}_s)$$

Where $\mathbf{A}_s$ = normalized/standardized matrix

Suppose,

1	3
2	5
6	4

Average = $\begin{matrix} 3 & 4 \end{matrix}$

S.Dev(σ) = $\begin{matrix} 2.65 & 1.00 \end{matrix}$

Then, $\mathbf{A}_s = (1/\sigma)(\mathbf{A} - \text{Ave Matrix})$

-0.76	-1.00
-0.38	1.00
1.13	0.00

So, $\mathbf{K} = [1/(n-1)] \mathbf{A}_s' \mathbf{A}_s$

1
2

X

-0.76	-0.38	1.13
-1.00	1.00	0.00

X

-0.76	-1.00
-0.38	1.00
1.13	0.00

where, n=samples = 3

1.00	0.19
0.19	1.00

K =

Multivariate Analysis

Multivariate Techniques & Matrix Operation

Models of Association in MVA

$$\mathbf{C} = \begin{matrix} & \begin{matrix} V_1 & V_2 & \dots & V_m \end{matrix} \\ \begin{bmatrix} \text{var}_{1,1} & \text{cov}_{1,2} & \dots & \text{cov}_{1,m} \\ \text{cov}_{2,1} & \text{var}_{2,2} & \dots & c_{2,m} \\ \vdots & \vdots & \vdots & \vdots \\ \text{cov}_{m,1} & \text{cov}_{m,2} & \dots & \text{var}_{m,m} \end{bmatrix} & \begin{matrix} V_1 \\ V_2 \\ \vdots \\ V_m \end{matrix} \end{matrix} \quad \mathbf{K} = \begin{matrix} & \begin{matrix} V_1 & V_2 & \dots & V_m \end{matrix} \\ \begin{bmatrix} 1 & k_{1,2} & \dots & k_{1,m} \\ k_{2,1} & 1 & \dots & k_{2,m} \\ \vdots & \vdots & \vdots & \vdots \\ k_{m,1} & k_{m,2} & \dots & 1 \end{bmatrix} & \begin{matrix} V_1 \\ V_2 \\ \vdots \\ V_m \end{matrix} \end{matrix}$$

Variance-Covariance Matrix

- Variance measures the scale of data spread around the mean value.
- Covariance (cov) determines the direction of change between two random variables.
- A positive cov means that the two variables are positively correlated and move in the same direction.
- A negative cov signifies that the two variables are inversely related and move in the opposite direction.

Correlation Matrix

- Correlation coefficient: is the term referring to resulting correlation measurement and it maintains a value between positive one and negative one.
- Correlation defines the direction and magnitude of how two variables vary with each other.
- A correlation of '+1' indicates that the two variables move in the same direction significantly.
- A correlation of '-1' indicates that the two variables move in the opposite direction significantly.

Multivariate Analysis

Eigen-structure

Linear Transformation

Any linear transformation can be represented in matrix form
=> transforming vectors into vectors (solving simultaneous linear equations):

$$\begin{aligned}x_1^* &= a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_m \\x_2^* &= a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_m \\&\vdots \\x_n^* &= a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_m\end{aligned}$$



Transformation Matrix

Images

$$\mathbf{x}^* = \mathbf{A}\mathbf{X}$$

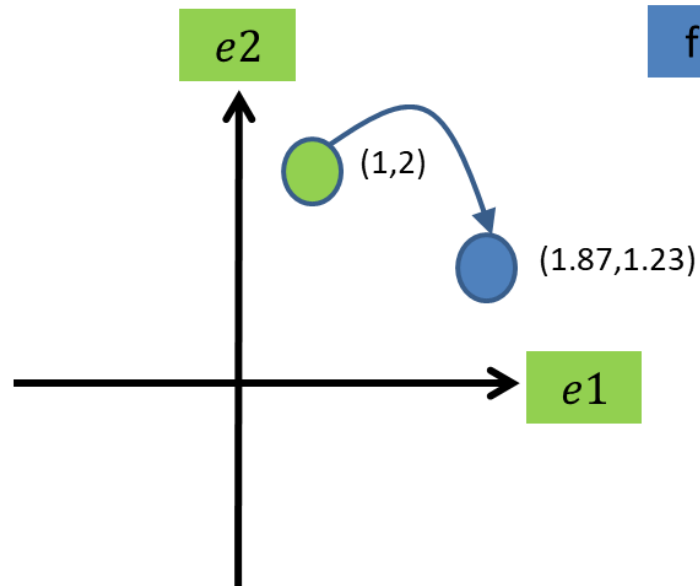
Pre-images

Multivariate Analysis

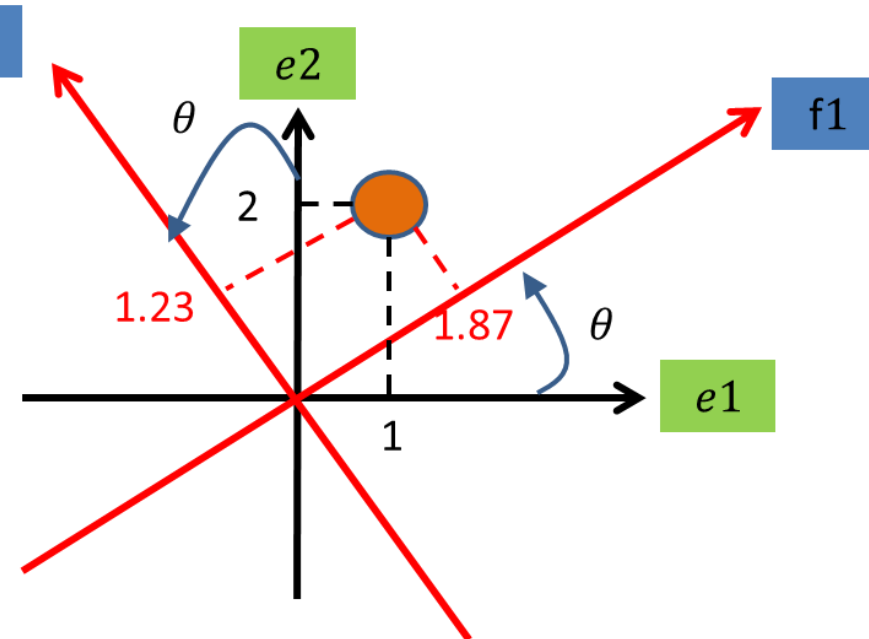
Eigen-structure

Geometrically, linear transformation usually involves rotation

Point rotation: the points move clockwise or counterclockwise while the basis vectors remained fixed.



Basis vector rotation: the original points remained fixed, while the basis vectors move clockwise or counterclockwise



Multivariate Analysis

Eigen-structure

Var-cov matrix or
correlation matrix

$$\mathbf{Ax} = \lambda \mathbf{x}$$

Eigenvectors = invariant vectors ($\mathbf{x}$), in which, those vectors that map into themselves or multiples of themselves under a given transformation map, $\mathbf{A}$.

Eigenvalues (characteristic roots) = a scalar (λ) corresponds to that particular eigenvectors.

$$\mathbf{Ax} = \mathbf{x}^*$$

$$\begin{bmatrix} -3 & 5 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 5 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

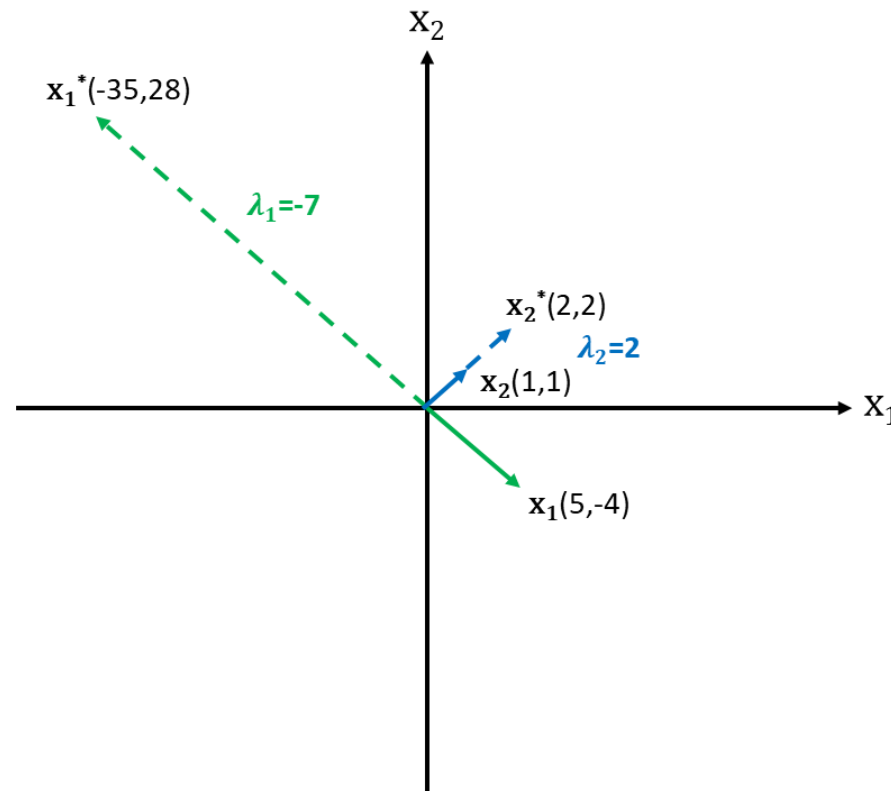
Eigenvalues, λ

Eigenvectors, $\mathbf{x}$

Multivariate Analysis

Eigen-structure

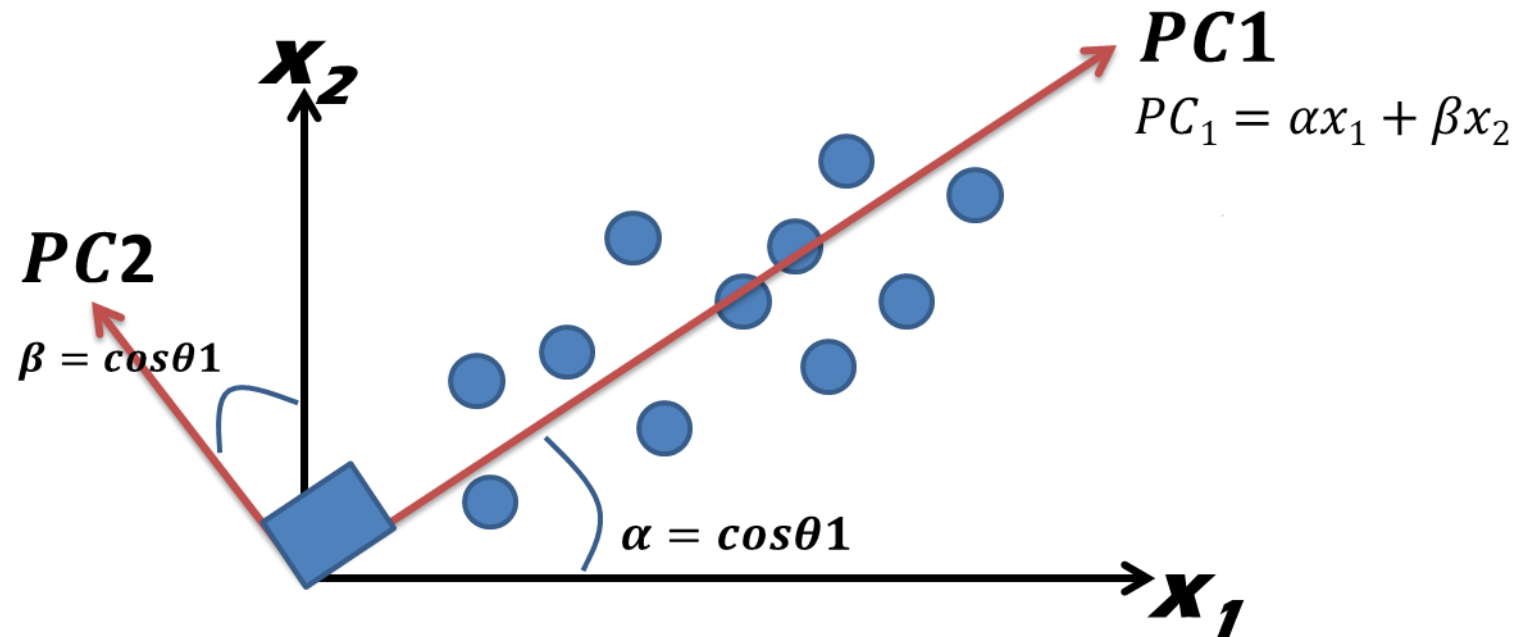
The geometrical illustration of eigenvectors and eigenvalues



Multivariate Analysis

Variance-covariance Transformation

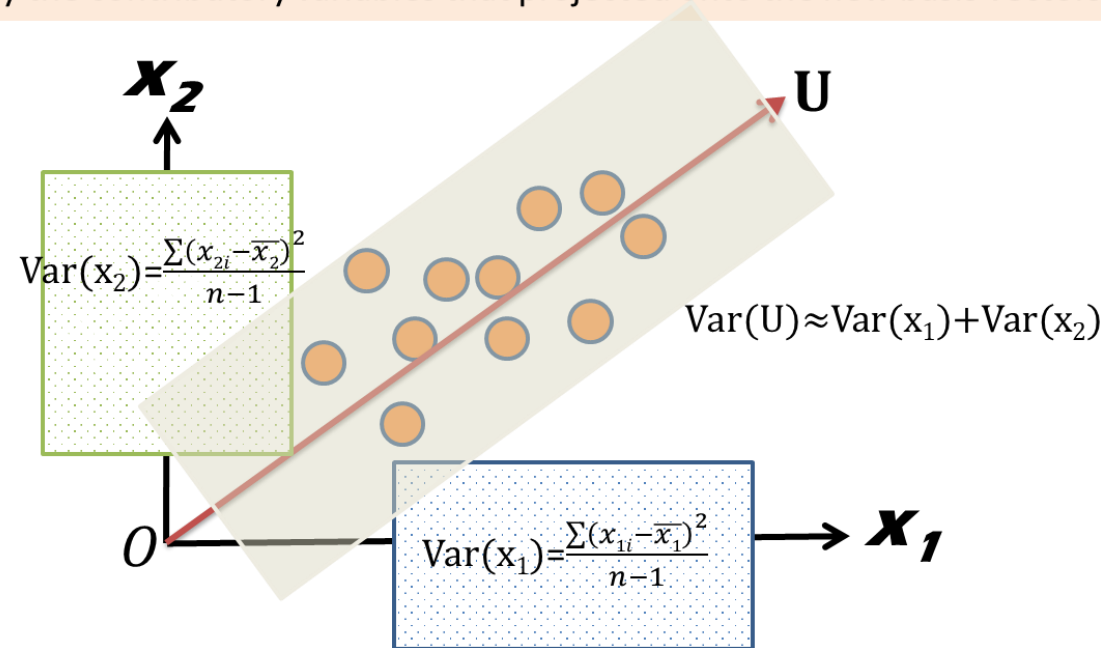
- MSPM framework depends on 'Multivariate Dimensional Reduction Technique' (variable interdependence analysis).
- The main aim = representing the objects of the investigation in terms of fewer dimensions (**PRINCIPAL COMPONENTS - PC**) than originally expressed:



Multivariate Analysis

Variance-covariance Transformation

- The **Optimal PC** is chosen such that the variance of this model is maximized under a set of orthogonal basis vectors, U .
- It means that the original samples are maximally separated along the linear composite, U according to the amount of original variance.
- This is an attempt of accounting for as much as possible of the original variation shared by the contributory variables that projected onto the new basis vectors, U .



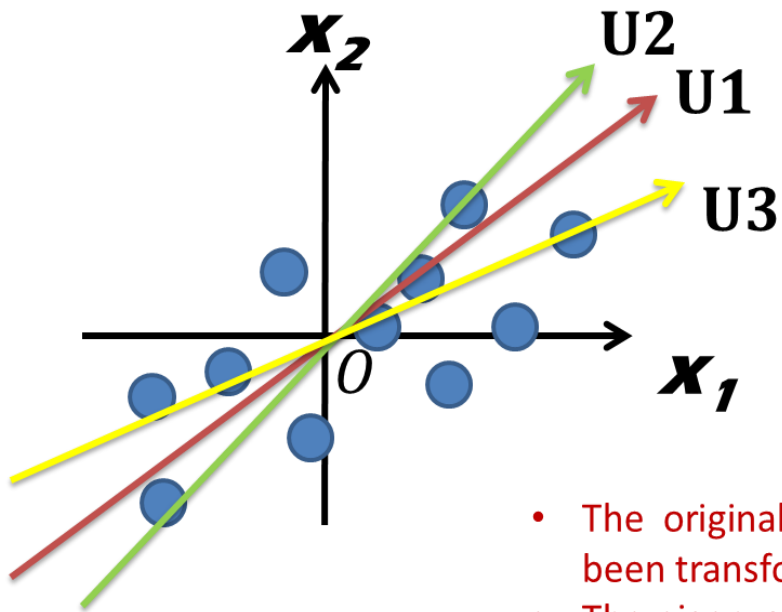
Multivariate Analysis

Variance-covariance Transformation

How to calculate the variance of the transformed PC variables?

$$PC_1 = u_{11}x_1 + u_{21}x_2 \quad PC_2 = u_{12}x_1 + u_{22}x_2 \quad PC_3 = u_{13}x_1 + u_{23}x_2$$

$$C(U1) = U1'C(X)U1 \quad C(U2) = U2'C(X)U2 \quad C(U3) = U3'C(X)U3$$



The variance measure of the newly transformed data can be obtained prior to transformation:

$$C(U) = U'C(X)U$$

Where, $C(U)$ = the new var-cov matrix (diagonal matrix of eigenvalue), $C(X)$ = the original var-cov matrix and U = eigenvectors

- The original variance-covariance matrix, $C(X)$ has been transformed into $C(Z)$ by U !
- The eigenvalue is the magnitude of variance which should be transformed from $C(X)$!

Important Properties:

- The sum of the main diagonal entries $C(U)$ is equal to the sum of the main diagonal entries of $C(X)$, whereby the 1st entry will always be the largest => maximally distributed according to the original variance.
- The off-diagonal entries are much smaller in absolute value than their counterpart entries in $C(X)$ => the transformed data are not correlated with each other.

Multivariate Analysis

Variance-covariance Transformation

- The rank of $\mathbf{A}$, denoted by $r(\mathbf{A})$, is defined as the maximum number of linearly independent rows (columns) => the maximum dimensions that can be compressed.
- If $r(\mathbf{A})=k$, then there exist k rows and k columns, where $k \leq \min(m, n)$.
- The number of nonzero or positive eigenvalues of larger than 1 of $\mathbf{A}$ is equal to its rank.

$\mathbf{C}(\mathbf{Z})=$

183	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	102	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	82	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	64	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	52	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	43	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	36	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	33	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	24	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	15	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

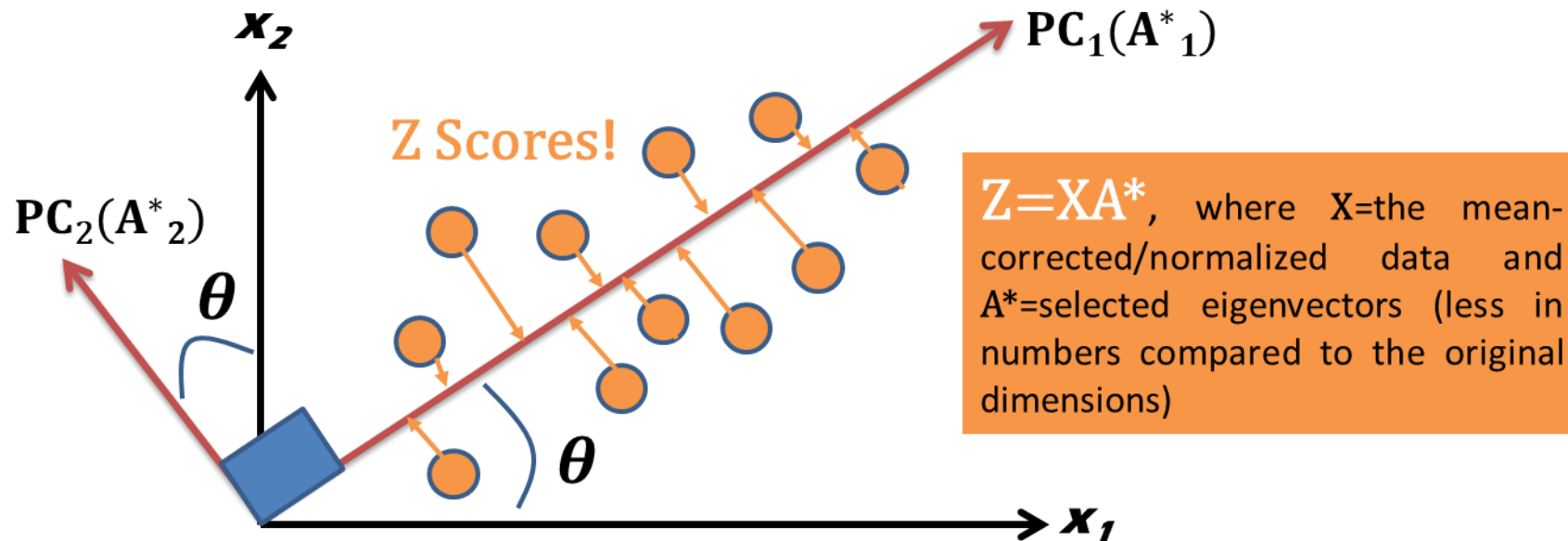
Define the matrix rank of the given $\mathbf{C}(\mathbf{Z})$!

- From the given $\mathbf{C}(\mathbf{Z})$ there are 13 different numbers of eigenvalues (diagonal entries) – there are 13 original variables from the raw data matrix!
- Only 10 numbers are larger than 1 (183, 102,...15) – matrix rank = 10!
- So, a maximum of 10 eigenvectors can be used as the new compressed dimensions (PCs) or new axis!

Multivariate Analysis

Variance-covariance Transformation

- Component scores (Z) are the projections of the original variables on each of the new axis (PCs) => the coordinates of the original variables that interpreted on the new dimensions of PCs!



- The variance of each column of Z will equal to its respective eigenvalues!

Exercise 1

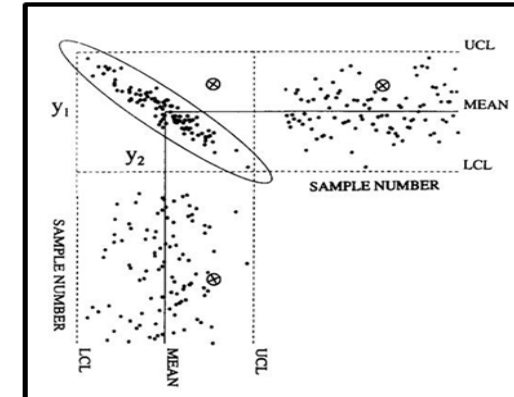
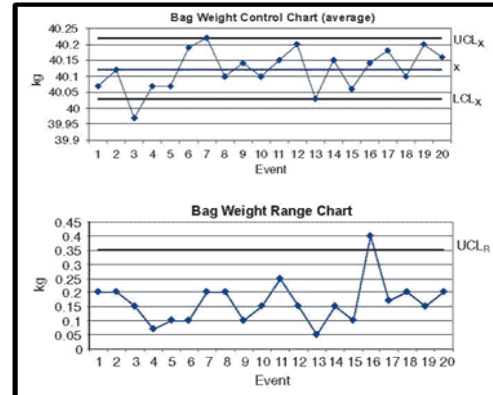
Explore the procedures of calculating the PCA scores using any of computational software that you familiar.

MSPM System

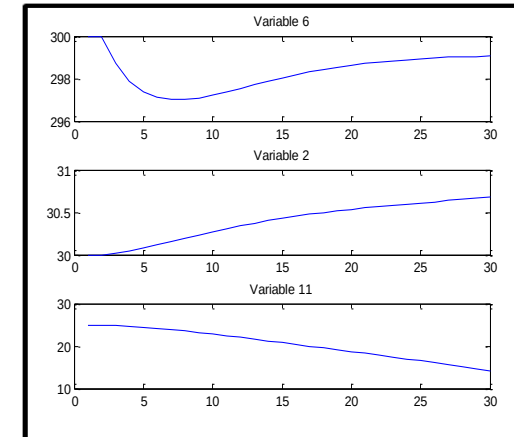
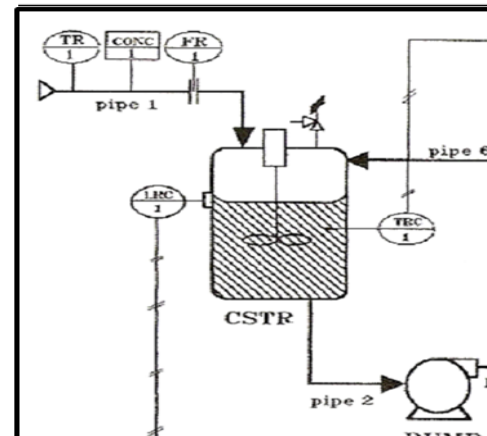
General Framework of MSPM

MOTIVATION: SPC suffers from 2 major drawbacks

1. SPC ignores the correlation among of inter-related variables.



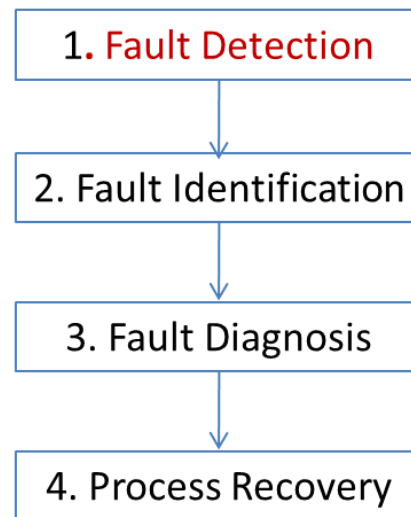
2. SPC does not provide the process/operation insight when problems occurred.



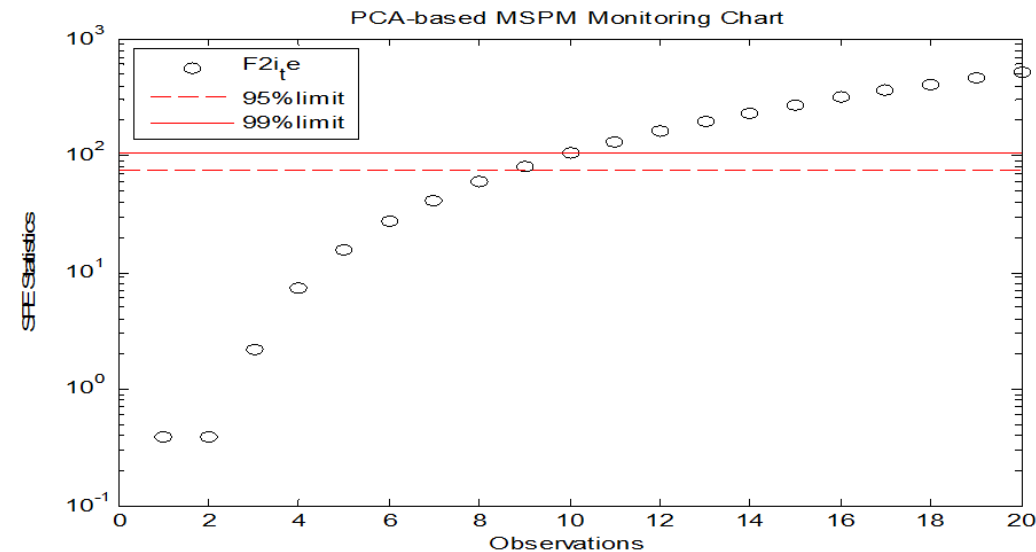
MSPM System

General Framework of MSPM

The **4 Main Steps** of MSPM System



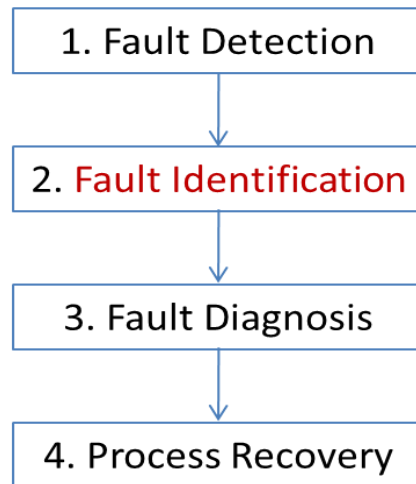
Fault detection: to designate the departure of observed samples from an acceptable range using a set of parameters.



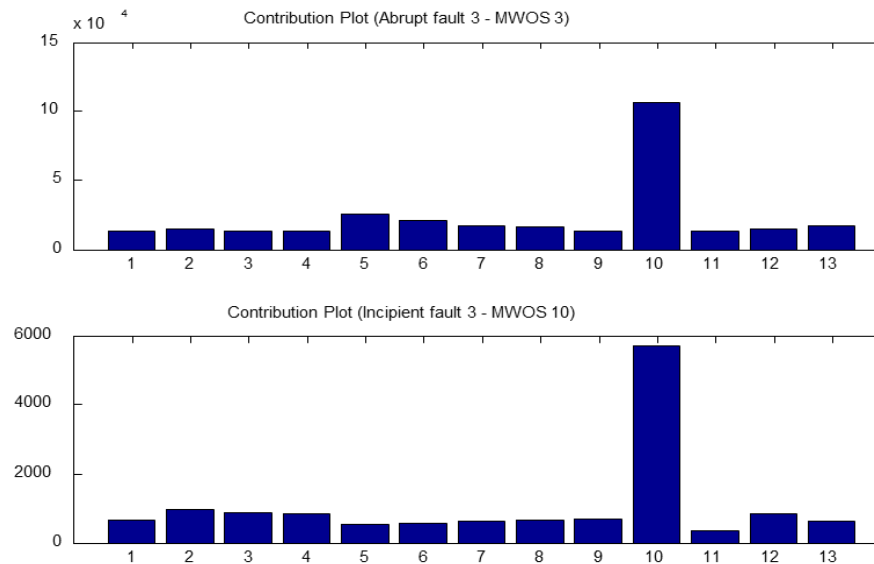
MSPM System

General Framework of MSPM

The 4 Main Steps of MSPM System



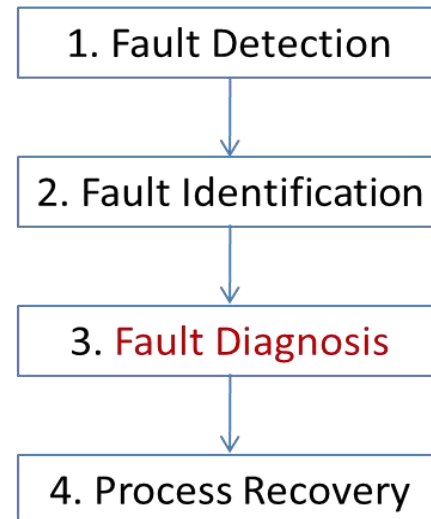
Fault identification: identifying the observed process variables that are most relevant to the fault which is typically identified by using the contribution plot technique.



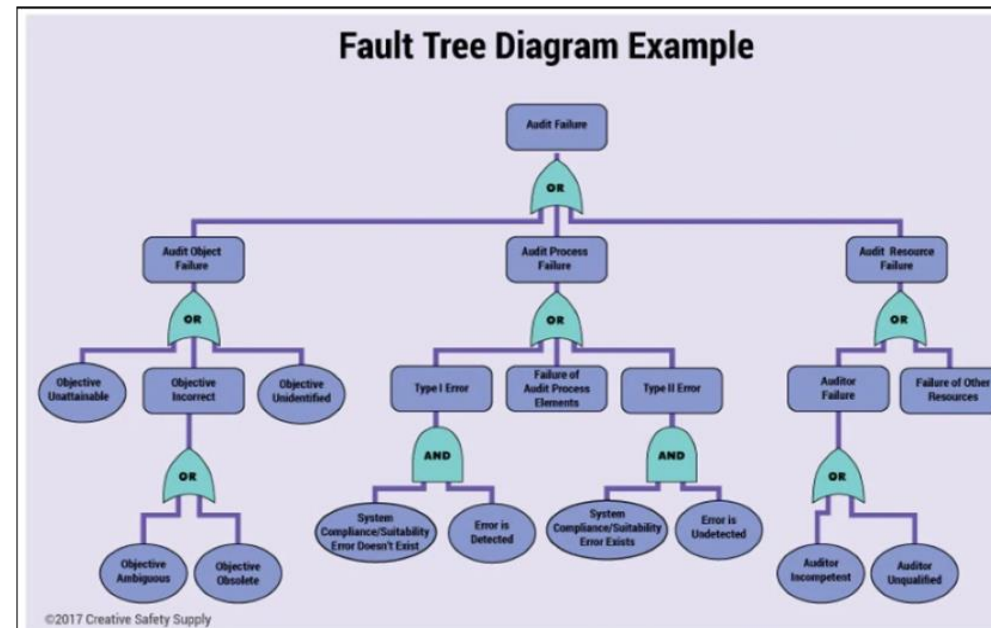
MSPM System

General Framework of MSPM

The 4 Main Steps of MSPM System



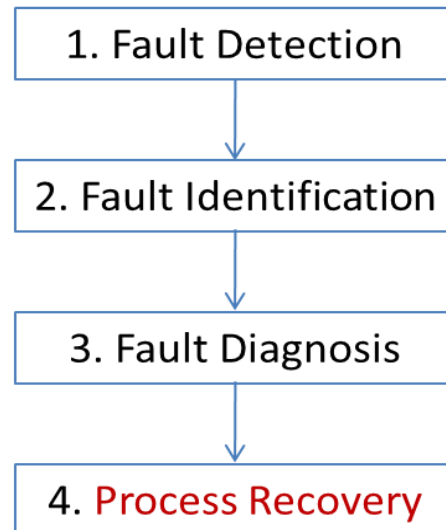
Fault diagnosis: specifically determines the type of fault which has been significantly (and should be also validated) contribute to the signal.



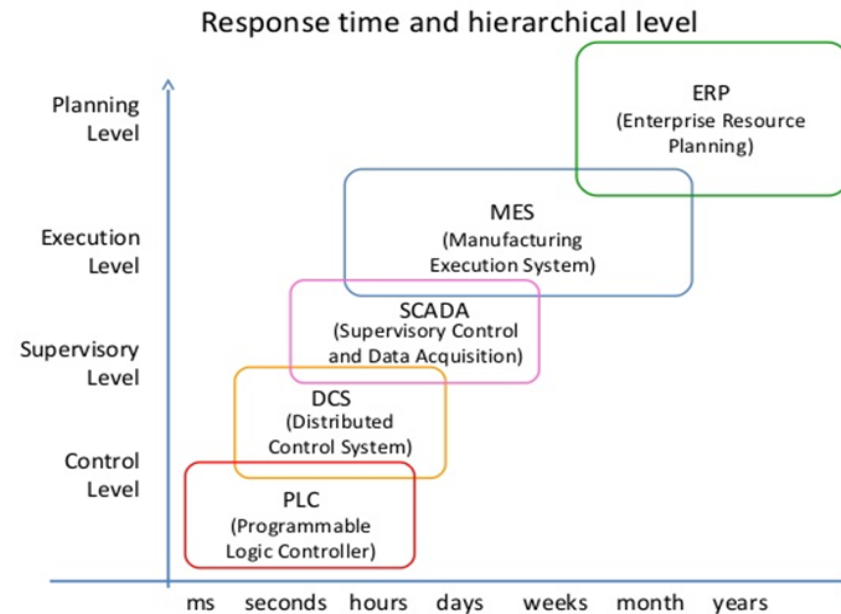
MSPM System

General Framework of MSPM

The 4 Main Steps of MSPM System



Process recovery: remove the cause(s) that contribute to the detected fault.

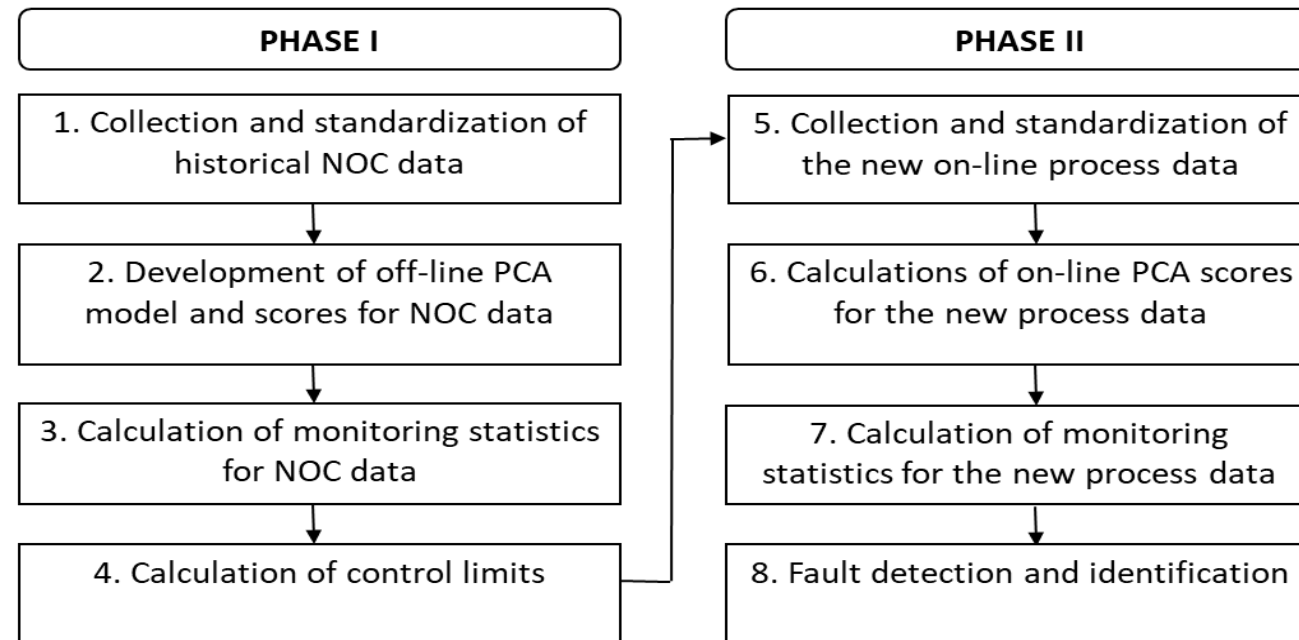


MSPM System

General Framework of MSPM

Fault Detection and Identification Procedures

The complete procedures of fault detection and identification in the MSPM framework comprise of two main phases - **off-line modelling and monitoring (phase I)** and **on-line monitoring (phase II)**:



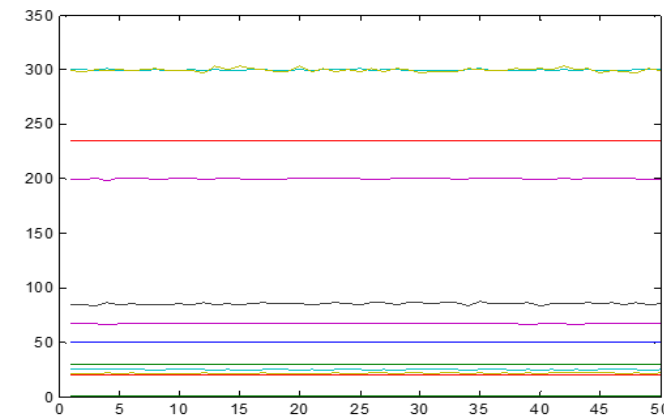
MSPM System

MSPM Development – Phase I

Fault Detection and Identification Procedures: Step 1

- Firstly, a set of normal operation condition (NOC) data, $\mathbf{X}_{n \times m}$ (n : samples, m : variables), are identified off-line based on the historical process data archive.
- NOC simply implies that the process is operated at the desired setting condition and produces satisfactory products that meet the qualitative as well as quantitative specified standard (Martin et al., 1996).

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,m} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,m} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{n,m} \end{bmatrix}$$



MSPM System

MSPM Development – Phase I

Fault Detection and Identification Procedures: Step 1

Then, the data are **standardized to zero mean and unit variance** with respective to each of the variables because PCA results depend on data scales.

$$\tilde{x}_{j,i} = \frac{(x_{j,i} - \bar{x}_i)}{\sigma_i}$$

Samples	Variables - Raw			Variables - Normalized		
	1	2	3	1	2	3
1	50.10	29.93	20.03	0.84	-1.21	0.44
2	50.09	30.07	20.05	0.77	0.87	0.56
3	50.15	29.98	19.89	1.63	-0.53	-0.46
4	50.07	30.02	19.83	0.44	0.05	-0.81
5	50.13	29.97	20.13	1.33	-0.59	1.05
6	49.95	29.98	19.78	-1.50	-0.46	-1.19
7	50.09	30.10	20.25	0.74	1.29	1.79
8	50.06	30.00	19.81	0.20	-0.14	-0.98
9	50.09	29.99	19.86	0.64	-0.38	-0.68
10	50.00	29.96	19.80	-0.79	-0.85	-1.05
11	50.07	29.96	19.83	0.33	-0.86	-0.87
12	49.89	30.17	20.11	-2.52	2.42	0.91
13	50.01	29.99	20.01	-0.61	-0.34	0.28
14	49.99	30.12	20.25	-0.80	1.68	1.81
15	50.04	30.04	20.15	-0.07	0.37	1.17
16	50.11	30.02	20.00	0.98	0.07	0.21
17	50.03	29.97	19.82	-0.20	-0.64	-0.93
18	50.02	29.96	19.79	-0.42	-0.80	-1.10
19	50.00	30.09	19.85	-0.71	1.18	-0.74
20	50.03	29.94	20.06	-0.28	-1.13	0.60
Average	50.04	30.01	19.96	0.00	0.00	0.00
S.Dev	0.063	0.066	0.158	1.000	1.000	1.000

MSPM System

MSPM Development – Phase I

Fault Detection and Identification Procedures: Step 2

In the second step, the development of PCA model for the NOC data requires the establishment of a set of **correlation (or variance-covariance) matrix, $C_{m \times m}$** .

$$C = \frac{1}{n-1} \tilde{X}' \tilde{X} = \begin{bmatrix} c_{1,1} & c_{1,2} & \cdots & c_{1,m} \\ c_{2,1} & c_{2,2} & \cdots & c_{2,m} \\ \vdots & \vdots & \vdots & \vdots \\ c_{m,1} & c_{m,2} & \cdots & c_{m,m} \end{bmatrix}$$

Number of
samples

Standardized NOC Data

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10
V1	1.00	-0.26	0.11	0.17	0.29	0.42	-0.36	0.01	-0.12	-0.24
V2	-0.26	1.00	0.21	-0.09	-0.26	-0.12	-0.04	0.18	-0.04	0.36
V3	0.11	0.21	1.00	-0.20	0.19	0.13	-0.15	0.18	-0.25	0.15
V4	0.17	-0.09	-0.20	1.00	-0.05	0.27	0.03	-0.15	-0.05	0.05
V5	0.29	-0.26	0.19	-0.05	1.00	0.25	-0.27	-0.06	0.15	0.02
V6	0.42	-0.12	0.13	0.27	0.25	1.00	-0.36	0.00	-0.19	0.29
V7	-0.36	-0.04	-0.15	0.03	-0.27	-0.36	1.00	-0.14	0.18	-0.26
V8	0.01	0.18	0.18	-0.15	-0.06	0.00	-0.14	1.00	-0.12	-0.09
V9	-0.12	-0.04	-0.25	-0.05	0.15	-0.19	0.18	-0.12	1.00	-0.12
V10	-0.24	0.36	0.15	0.05	0.02	0.29	-0.26	-0.09	-0.12	1.00

MSPM System

MSPM Development – Phase I

Fault Detection and Identification Procedures: Step 2

C is then transformed into a set of basic structures of eigen-based formula =>

$$C = V \Lambda V^T$$

Transposed Eigenvectors Matrix

Eigenvectors Matrix

-0.31	-0.18	-0.14	0.40	0.01	0.00	0.13	-0.50	-0.27	0.59
0.08	0.45	0.19	-0.26	-0.14	0.25	0.32	-0.38	-0.55	-0.23
-0.15	0.21	0.35	0.11	-0.54	-0.28	0.14	-0.33	0.54	-0.07
-0.08	-0.05	-0.52	-0.12	0.04	0.05	0.79	0.09	0.26	-0.07
-0.30	-0.41	0.33	-0.18	-0.14	-0.01	0.14	0.17	-0.14	-0.03
-0.41	0.06	-0.34	-0.09	-0.26	0.16	-0.28	0.00	-0.01	-0.17
0.39	-0.19	-0.18	-0.09	-0.51	-0.02	-0.03	0.05	-0.04	0.20
-0.03	0.22	0.22	0.42	-0.17	0.67	0.12	0.42	0.10	0.20
0.12	-0.33	0.10	-0.32	0.19	0.58	-0.12	-0.46	0.40	0.02
-0.15	0.36	0.03	-0.60	0.02	-0.07	-0.06	0.18	0.10	0.66

Eigenvalues Matrix

183	0	0	0	0	0	0	0	0	0
0	102	0	0	0	0	0	0	0	0
0	0	82	0	0	0	0	0	0	0
0	0	0	64	0	0	0	0	0	0
0	0	0	0	52	0	0	0	0	0
0	0	0	0	0	43	0	0	0	0
0	0	0	0	0	0	36	0	0	0
0	0	0	0	0	0	0	33	0	0
0	0	0	0	0	0	0	0	24	0
0	0	0	0	0	0	0	0	0	15

MSPM System

MSPM Development – Phase I

Fault Detection and Identification Procedures: Step 2

The following equation presents a measure proportion of data variations captured by the first ' k_a ' principal components.

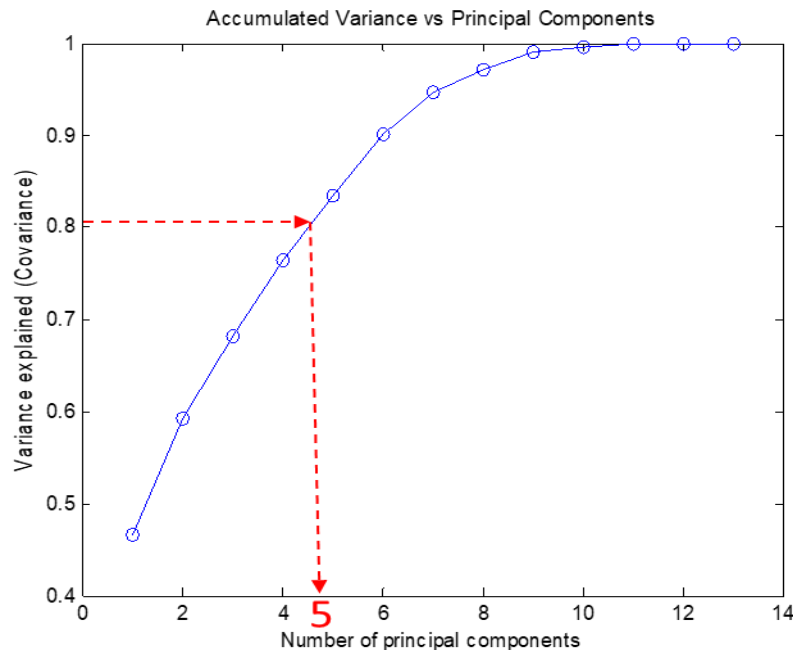
$$k_a = \frac{\lambda_1 + \lambda_2 + \dots + \lambda_a}{\lambda_1 + \lambda_2 + \dots + \lambda_a + \dots + \lambda_m}$$

$$k_1 = \frac{183}{(183 + 102 + \dots + 15)}$$

$$k_2 = \frac{183 + 102}{(183 + 102 + \dots + 15)}$$

$\vdots$

$$k_m = \frac{183 + 102 + \dots + 15}{(183 + 102 + \dots + 15)}$$



MSPM System

MSPM Development – Phase I

Fault Detection and Identification Procedures: Step 2

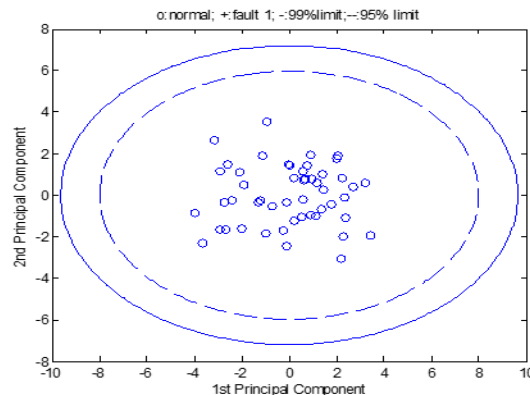
Finally, the PCA model can be simply developed by this equation:

$$\mathbf{P} = \mathbf{\tilde{X}} \mathbf{V}^a$$

Standardized Data
Selected Eigenvectors

$$\mathbf{P} = [\mathbf{p}_1 \ \cdots \ \mathbf{p}_a]$$

$$= \begin{bmatrix} \tilde{x}_{1,1}v_{1,1} + \cdots + \tilde{x}_{1,m}v_{m,1} & \cdots & \tilde{x}_{1,1}v_{1,a} + \cdots + \tilde{x}_{1,m}v_{m,a} \\ \vdots & \cdots & \vdots \\ \tilde{x}_{n,1}v_{1,1} + \cdots + \tilde{x}_{n,m}v_{m,1} & \cdots & \tilde{x}_{n,1}v_{1,a} + \cdots + \tilde{x}_{n,m}v_{m,a} \end{bmatrix}$$



Samples	PC				
	1	2	3	4	5
1	-1.3	-0.37	-0.21	-1.17	-1.92
2	0.21	-1.24	0.02	-0.39	-0.76
3	-2.91	1.15	1.08	0.11	-1.43
4	2.37	-1.11	-3.06	-0.15	-0.59
5	-2.72	-0.35	1.46	-0.97	0.11
6	1.45	0.23	-0.02	0.95	-0.69
7	-2.93	-1.65	2.86	-0.57	0.87
8	-1	-1.85	-2.5	-0.64	-0.4
9	-1.24	-0.24	1.34	-1.05	-1.39
10	0.63	0.71	0.1	0.74	-1.18
11	-1.12	1.91	0.12	1.28	-1.31
12	3.46	-1.93	1.25	1.91	0.77
13	-3.99	-0.88	-1.42	1.93	-0.27
14	-0.1	-2.44	1.2	-0.02	1.87
15	-2.68	-1.67	0.26	-0.11	0.61
16	-0.71	-0.54	-1.72	-1.25	1.05
17	0.58	1.13	-1.34	-1	0.54
18	1.17	0.6	-0.27	-1.31	-1.68
19	0.68	0.77	2.16	-0.28	-0.35
20	-1.91	0.48	-1.62	-0.32	1.35

MSPM System

MSPM Development – Phase I

Fault Detection and Identification Procedures: Step 3

The third step basically involves calculation of the **Hotelling's T^2** and SPE monitoring statistics.

Samples	PC				
	1	2	3	4	5
1	-1.3	-0.37	-0.21	-1.17	-1.92
2	0.21	-1.24	0.02	-0.39	-0.76
3	-2.91	1.15	1.08	0.11	-1.43
4	2.37	-1.11	-3.06	-0.15	-0.59
5	-2.72	-0.35	1.46	-0.97	0.11
6	1.45	0.23	-0.02	0.95	-0.69
7	-2.93	-1.65	2.86	-0.57	0.87
8	-1	-1.85	-2.5	-0.64	-0.4
9	-1.24	-0.24	1.34	-1.05	-1.39
10	0.63	0.71	0.1	0.74	-1.18
11	-1.12	1.91	0.12	1.28	-1.31
12	3.46	-1.93	1.25	1.91	0.77
13	-3.99	-0.88	-1.42	1.93	-0.27
14	-0.1	-2.44	1.2	-0.02	1.87
15	-2.68	-1.67	0.26	-0.11	0.61
16	-0.71	-0.54	-1.72	-1.25	1.05
17	0.58	1.13	-1.34	-1	0.54
18	1.17	0.6	-0.27	-1.31	-1.68
19	0.68	0.77	2.16	-0.28	-0.35
20	-1.91	0.48	-1.62	-0.32	1.35

$$T_i^2 = \sum_{j=1}^a \frac{p_{i,j}^2}{\lambda_j}$$

$$T_1^2 = \frac{(-1.3)^2}{183} + \frac{(-0.37)^2}{102} + \dots + \frac{(-1.92)^2}{52}$$

$$T_{20}^2 = \frac{(-1.91)^2}{183} + \frac{(0.48)^2}{102} + \dots + \frac{(1.35)^2}{52}$$

Hotelling's T^2 conceptually shows the **magnitude of deviation from the centre** of multivariate data

MSPM System

MSPM Development – Phase I

Fault Detection and Identification Procedures: Step 3

The third step basically involves calculation of the Hotelling's T^2 and **Squared Prediction Errors (SPE)** monitoring statistics.

$$\tilde{\mathbf{E}} = \tilde{\mathbf{X}} - \hat{\mathbf{X}}$$

$$= \tilde{\mathbf{X}} - \mathbf{P}_a \mathbf{V}_a^T$$

Where, $\tilde{\mathbf{X}}$ is the standardized NOC data
and, $\hat{\mathbf{X}}$ is the predicted standardized NOC data

$$SPE_i = \tilde{\mathbf{e}}_i \tilde{\mathbf{e}}_i^T$$

Transposed Selected Eigenvectors
Multivariate Scores

Suppose $\tilde{\mathbf{E}}$ is the squared error matrix, then

$$\mathbf{E} = \begin{bmatrix} x_{1,a} & x_{1,b} & \cdots & x_{1,m} \\ x_{2,a} & x_{2,b} & \cdots & x_{2,m} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n,a} & x_{n,b} & \cdots & x_{n,m} \end{bmatrix}$$

$SPE_1 = x_{1a}^2 + x_{1b}^2 + \cdots + x_{1m}^2$
 $\vdots$
 $SPE_n = x_{na}^2 + x_{nb}^2 + \cdots + x_{nm}^2$

SPE conceptually measures the
**consistency of the correlation among
of the variables that analyzed**

MSPM System

MSPM Development – Phase I

Fault Detection and Identification Procedures: Step 4

The final task in phase I (4th step) deals with developing the control limits for both of the statistics.

T^2 limits ($\alpha = 95\%$ or 99% confidence limits) $\rightarrow T_{\alpha} = \frac{A(n-1)}{(n-A)} F_{A, n-A, \alpha}$

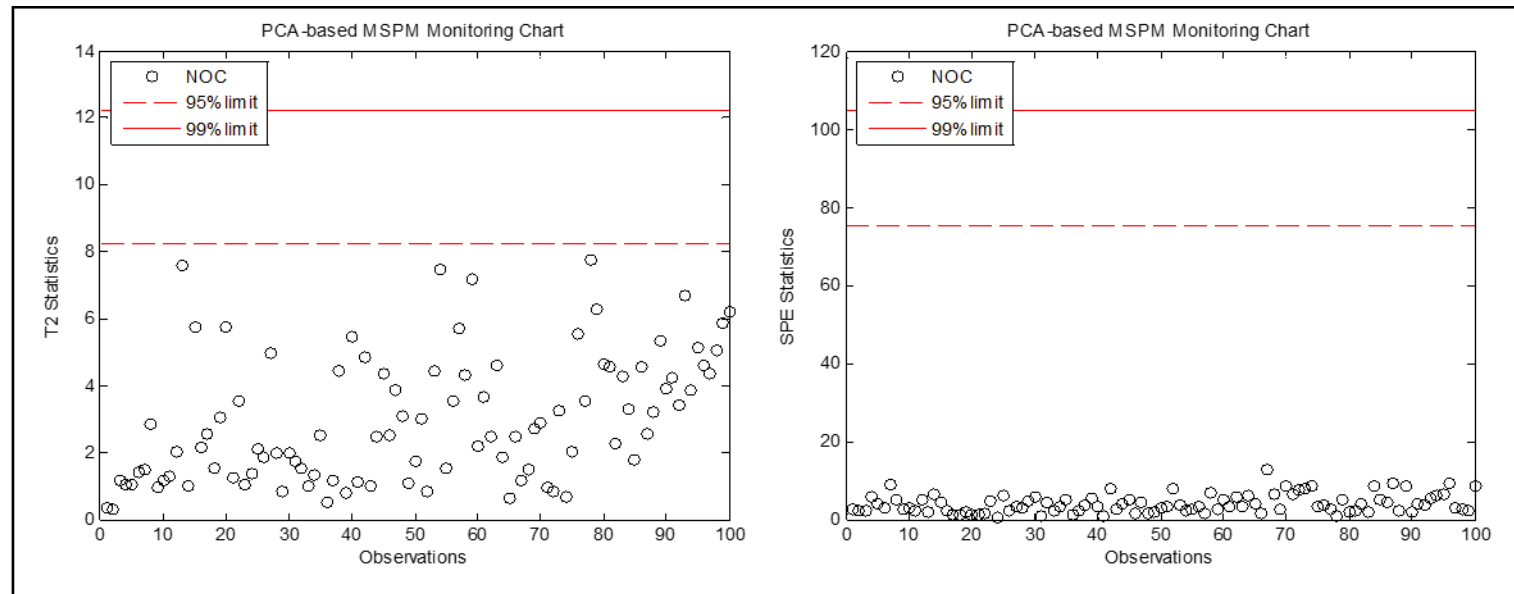
SPE limits ($\alpha = 95\%$ or 99% confidence limits) $\rightarrow SPE_{\alpha} = \theta_1 \left(\frac{z_{\alpha} \sqrt{2\theta_2 h_0^2}}{\theta_1} + \frac{\theta_1 h_0 (h_0 - 1)}{\theta_1^2} + 1 \right)^{\frac{1}{h_0}}$

MSPM System

MSPM Development – Phase I

Fault Detection and Identification Procedures: Step 4

T^2 and SPE progressions of NOC data!

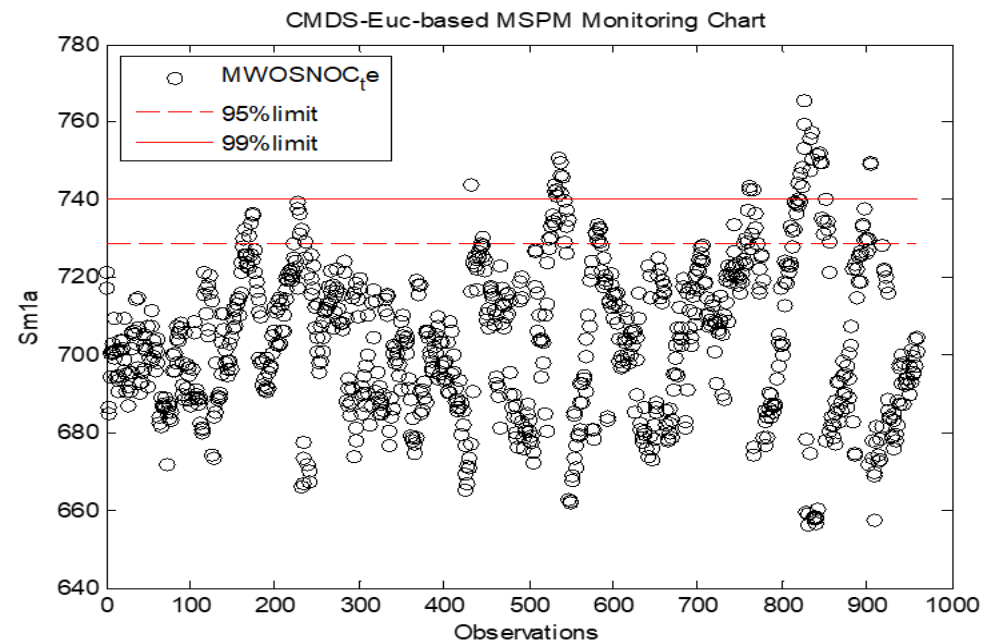


MSPM System

MSPM Development – Phase I

Phase I - False Alarm Rates (FAR)

- False alarm rate (FAR) analysis is conducted to evaluate the robustness of the monitoring statistics
- High FAR indicates that the developed limit settings are unsuitable to be used for monitoring (another set of NOC data is required to build up the new monitoring limits)



$$FAR = \frac{\text{Total amount of statistics in the NOC set located beyond the 99\% limit}}{\text{Total amount of NOC samples}}$$

MSPM System

MSPM Development – Phase II

Fault Detection and Identification Procedures: Step 5

- Step 5 follows exactly the same procedure of Step 1 in Phase I.
- However, all the averages and standard deviations of Phase I are utilized again to normalize the Case Data of Phase II (no need to find the new averages and s. deviations from the Case Data).

CASE DATA - RAW

1	2	3	4	5	6	7	8	9	10
50.00	30.00	20.00	300.00	200.00	300.00	85.72	0.11	0.80	234.50
50.00	30.00	20.00	300.00	200.00	300.00	85.72	0.11	0.80	234.50
49.05	29.91	20.18	295.22	199.80	300.84	100.02	0.11	0.80	234.47
49.27	29.98	20.06	294.65	199.87	296.53	99.81	0.11	0.80	234.34
49.27	29.78	20.01	294.50	200.32	297.06	97.62	0.11	0.80	234.30
49.35	29.91	20.04	295.39	199.23	295.76	97.19	0.11	0.80	234.34
49.39	29.88	20.06	295.87	199.25	296.41	95.30	0.11	0.80	234.33
49.39	29.82	20.24	294.52	200.04	297.90	94.59	0.11	0.80	234.28
49.53	30.01	20.03	294.25	200.07	295.07	93.56	0.11	0.79	234.30
49.53	29.99	20.20	296.01	199.75	294.19	93.55	0.11	0.80	234.38



$$\tilde{x}_{j,i}[Case] = \frac{(x_{j,i}[Case] - \bar{x}_i[NOC])}{\sigma_i[NOC]}$$

CASE DATA - NORMALIZED

1	2	3	4	5	6	7	8	9	10
0.48	-0.58	0.34	0.11	0.51	0.06	0.22	0.81	-0.01	-0.58
0.48	-0.58	0.34	0.11	0.51	0.06	0.22	0.81	-0.01	-0.58
-0.67	-0.68	1.78	-8.09	0.31	0.61	6.16	2.80	0.69	-0.61
-0.40	-0.60	0.78	-9.07	0.38	-2.20	6.08	1.80	-0.36	-0.74
-0.39	-0.82	0.41	-9.32	0.83	-1.86	5.16	2.80	-1.40	-0.78
-0.30	-0.68	0.65	-7.80	-0.26	-2.70	4.98	1.80	0.34	-0.74
-0.26	-0.71	0.83	-6.97	-0.23	-2.28	4.20	-0.18	1.04	-0.75
-0.26	-0.78	2.21	-9.28	0.56	-1.30	3.90	1.80	-1.05	-0.80
-0.09	-0.57	0.61	-9.75	0.58	-3.15	3.47	0.81	-3.15	-0.78
-0.08	-0.59	1.92	-6.72	0.27	-3.72	3.47	-0.18	1.04	-0.71

MSPM System

MSPM Development – Phase II

Fault Detection and Identification Procedures: Step 6

- Step 6 follows exactly the same procedure of Step 2 in Phase I.
- However, the scores are computed directly by using the eigenvectors of Phase I (no need to find the new correlation matrix, eigenvectors or eigenvalues from the Case Data)

$$\mathbf{P}_{\text{Case}} = \mathbf{\tilde{X}}_{\text{Case}} \mathbf{V}_a$$

Standardized Case Data Selected Eigenvectors of Phase I

PC1	PC2	PC3
-1.13	-0.51	-0.77
-1.13	-0.51	-0.77
-1.16	0.37	0.01
-1.23	1.58	0.01
-1.05	3.38	0.35
1.14	22.52	-0.36
1.33	24.95	-0.73
1.72	27.56	-0.62
1.90	29.98	-0.95
2.27	32.56	-0.78

MSPM System

MSPM Development – Phase II

Fault Detection and Identification Procedures: Step 7

- Step 7 follows exactly the same procedure of Step 3 in Phase I.
- However, the T^2 and SPE statistics of the Case Data are computed directly by using the eigenvectors as well as eigenvalues of Phase I

$$T_{Case_i}^2 = \sum_{j=1}^a \frac{p_{i,j}^2}{\lambda_j}$$

PC Scores of Case Data ($\mathbf{P}_{case}$)

Corresponding eigenvalues of Phase I

$$SPE_{Case_i} = x_{ia}^2 + x_{ib}^2 + \dots + x_{im}^2$$

$$\tilde{\mathbf{E}}_{Case} = \begin{bmatrix} x_{1,a} & x_{1,b} & \dots & x_{1,m} \\ x_{2,a} & x_{2,b} & \dots & x_{2,m} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n,a} & x_{n,b} & \dots & x_{n,m} \end{bmatrix}$$

$$\begin{aligned} \text{Where } \tilde{\mathbf{E}}_{Case} &= \tilde{\mathbf{X}}_{Case} - \hat{\mathbf{X}}_{Case} \\ &= \tilde{\mathbf{X}}_{Case} - \mathbf{P}_{Case} \mathbf{V}_a^T \end{aligned}$$

MSPM System

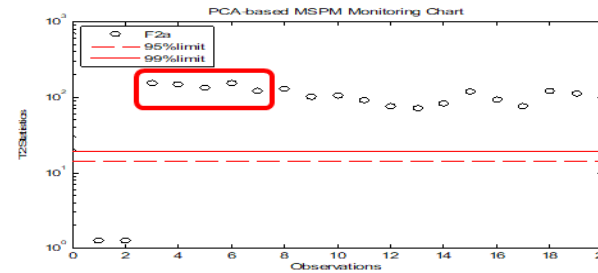
MSPM Development – Phase II

Fault Detection and Identification Procedures: Step 8

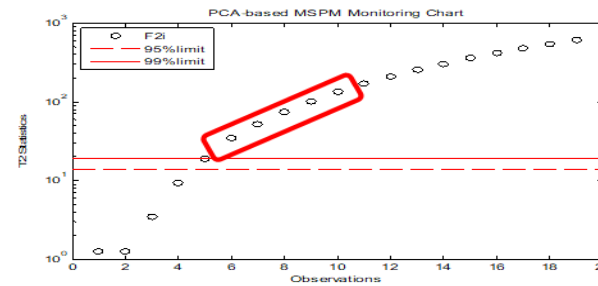
Step 8 conducts the fault detection and identification operation.

Fault detection:

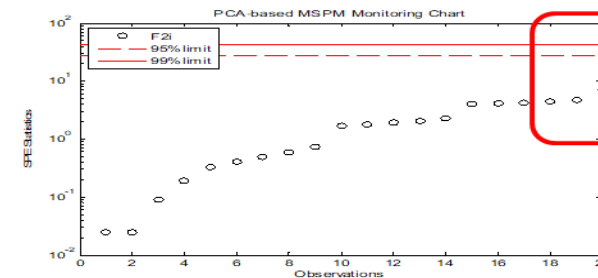
- A fault situation is regarded as a result of an occurrence of a special event that is not in conformance to the common cause nature .
- Technically, a fault situation will be declared if either of the monitoring statistics exceeding its respective control limit for a pre-defined successive number of samples consistently (at least 5 samples).



Abrupt Fault –
quick detection



Incipient Fault –
slow detection



Incipient Fault –
missed detection

MSPM System

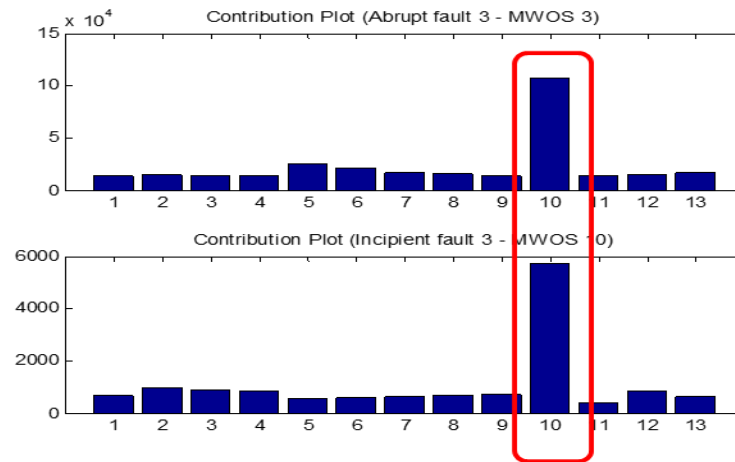
MSPM Development – Phase II

Fault Detection and Identification Procedures: Step 8

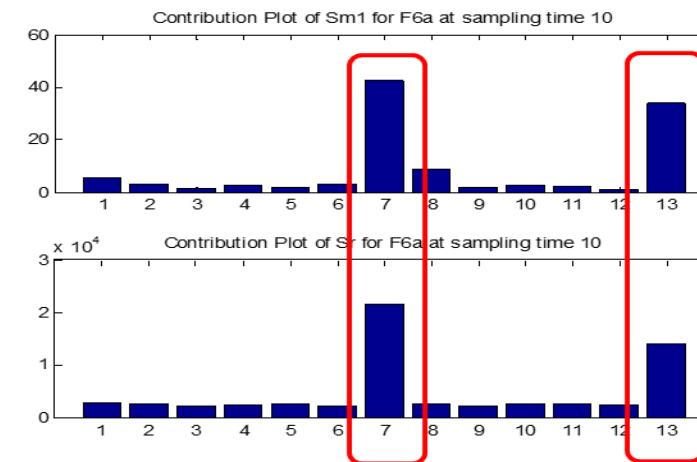
Step 8 conducts the fault detection and identification operation.

Fault Identification –

- The contribution plot technique is proposed to identify the potential set of variables that possibly connected to the detected problem.
- The contribution plot is developed corresponding to the time where any of the monitoring statistics firstly crossing beyond the monitoring limits consistently.



A single root cause – fault is identified & diagnosed.



Multiple root causes – conduct fault diagnosis.

MSPM System

MSPM Development – Phase II

Phase II – Fault Detection Sampling Time (FDsT) and Fault Detection Delayed Time (FDdT)

- FDsT is the time corresponding to the first instance of the monitoring statistics crosses the limit consistently.
- FDdT is the time duration between FDsT and the fault signal introduced into the system

Fault Detection Sampling Time		
Fault Cases	PCs 5	
	T2	SPE
1a	3	6
2a	3	x
3a	3	3
4a	3	3
5a	3	3
6a	3	3
7a	x	4
8a	3	3
9a	3	3
10a	3	3
11a	3	3



Fault Detection Sampling Time (Summary)	
1a	3
2a	3
3a	3
4a	3
5a	3
6a	3
7a	4
8a	3
9a	3
10a	3
11a	3



Fault Detection Delayed Time	
1a	1
2a	1
3a	1
4a	1
5a	1
6a	1
7a	2
8a	1
9a	1
10a	1
11a	1

Fault signal is introduced at sampling time 2

$$FDdT = FDsT - Ts$$

Where, Ts = the time of the signal introduced into the system.

Exercise 2

1. **Develop the MSPM procedures based on the specified NOC & CASE data of CSTR system that provided.**
2. **Develop and determine the appropriateness of using the MSPM system for process enhancement in the BPC plant operation.**